

Hype Check: Challenging Causal Foundation Models for Uplift

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Abstract

Causal foundation models such as CausalPFN estimate conditional average treatment effects (CATE) in context, without per-dataset training, and report strong results on academic benchmarks. Uplift modeling, however, is mainly a ranking problem: the goal is to decide which units to treat under a budget, not to recover each effect exactly. It is therefore unclear whether this advantage holds on real campaigns under a modest compute budget. We first compare an untuned CausalPFN with strong tuned baselines on ranking quality, and we test whether effect accuracy predicts it. We then identify the regimes where the advantage disappears: data beyond the context window, a small control group, and low conversion. Finally, we propose CausalPFN-Rank, a training-free adaptation that improves the in-context sample and re-ranks the model outputs, and we measure how much ranking quality it recovers. All comparisons are relative, use bootstrap confidence intervals, and cover both real randomized trials and semi-synthetic data.

Index Terms: Uplift modeling, conditional average treatment effect, causal foundation models, in-context learning, ranking evaluation.

1 Introduction

Marketing and retention decisions depend on uplift, the incremental effect of an action on an individual, formalized as the conditional average treatment effect (CATE). In practice the exact effect value matters less than a good ordering of units by incremental response, so that a limited budget reaches the most persuadable customers first. Ranking quality is therefore measured with the Qini coefficient and the area under the uplift curve, not with pointwise error.

Causal foundation models are transformers pre-trained on large collections of simulated causal models. They estimate CATE in context, with no task-specific training, and report strong benchmark results [2, 9, 18]. The most mature open model, CausalPFN, ships with pre-trained weights, a simple estimator interface, and calibrated uncertainty [2]. Recent evidence, though, shows that benchmark accuracy and real ranking quality can diverge. A large-scale study reports a plain S-Learner topping the Qini ranking among classical methods [21], and a benchmark on field-collected data finds most CATE estimates no more accurate than a constant-effect predictor [23].

We ask whether causal foundation models improve uplift ranking under realistic constraints, where they break, and whether a lightweight fix repairs them. We propose CausalPFN-Rank, a training-free adaptation that selects a better in-context sample and re-ranks the model outputs for the ranking objective. The project has three parts: investigate the model, find where it breaks, and fix it.

1.1 Hypotheses

- **H1:** Without tuning, CausalPFN matches strong tuned baselines on semi-synthetic effect accuracy, but does not beat them on ranking on real campaigns.
- **H2:** Ordering methods by effect accuracy (PEHE) agrees only weakly with ordering them by uplift (Qini), so the most accurate estimator is often not the one that ranks units best.
- **H3:** The advantage disappears in three regimes: data beyond the context window, a small control group, and low conversion.
- **H4:** Engineered context, balancing treated and control and averaging over several retrieved samples, recovers much of the ranking lost in the scale and small-control regimes, with no training.
- **H5:** Re-ranking the outputs improves uplift even when effect accuracy is unchanged, narrowing the gap to specialist methods.

1.2 Goals

- **G1:** Build a reproducible uplift-evaluation harness wrapping CausalPFN and the baselines over the dataset pool, and reproduce CausalPFN’s own uplift example as an early feasibility check.
- **G2:** Map the regimes where the advantage breaks by controlled subsampling, reporting uplift metrics with bootstrap confidence intervals, effect accuracy where ground truth exists, and the correlation between the two rankings.
- **G3:** Build and ablate CausalPFN-Rank, its context part and its output part, and report how much ranking each part recovers, with compute cost.

1.3 Significance of Hypotheses Confirmation

If validated:

- *Practical relevance:* a rule for when an untuned foundation model is enough for ranking, and when a tuned classical method or a specialist is worth the effort under a fixed budget.
- *Methodological value:* evaluation of causal foundation models shifts toward the decision objective rather than effect accuracy, and the post-processing recipe applies to any in-context causal model.
- *Reusable benchmark:* the harness and the map of where the model helps or breaks stay useful for a fast-moving area.

1.4 Validation Protocol

Datasets: The pool is small and chosen to span the regimes of interest, not to maximize count. Real randomized data without ground-truth effects: Criteo Uplift and X5 RetailHero at large scale [7], and Hillstrom, which is small and low-conversion [13]. Semi-synthetic data with ground-truth effects: IHDP [12] and

ACIC 2016 [8]. We subsample large tables for the scale axis and re-sample for the control-share and conversion axes.

Metrics:

- Ranking: the Qini coefficient and uplift at the top fraction, with bootstrap confidence intervals over at least ten seeds.
- Accuracy: PEHE against ground-truth effects on semi-synthetic data.
- Agreement: Spearman correlation between the PEHE-based and Qini-based rankings of the methods.
- Efficiency, logged separately: GPU-hours and inference time.

Baselines: The tuned classical core is the S-, T-, X- and DR-Learners on gradient boosting, plus the Causal Forest [1, 3, 15–17, 22]. We add the Bayesian Causal Forest and the neural learners DragonNet, TARNet, CFR and DESCN [5, 11, 19, 20, 24]. Two 2026 specialists run on their own regime: the Q-Learner for low conversion [10] and GP-CATE for a small control group [4]. CausalPFN is the foundation anchor [2].

2 Background & Literature Review

Under the potential-outcomes framework, the CATE is the expected difference in outcome from treating an individual with given covariates. In a randomized trial, treatment is independent of covariates, so the effect is the gap between the treated and control response surfaces and uplift can be validated directly. The task is to rank units by effect and treat a top fraction under budget, so ranking quality is summarized by the Qini coefficient and the area under the uplift curve [6]. This is why a model with worse point-wise accuracy can still rank better.

Meta-learners reduce effect estimation to supervised learning: the S-, T- and X-Learners [16], the orthogonal R-Learner [17], and doubly robust DR-Learners [15], here all on gradient boosting. Causal Forest and generalized random forests [1, 22], and Bayesian Causal Forest [11], strong on semi-synthetic benchmarks, add mature baselines with uncertainty. Representation-learning networks form the neural family [19, 20, 24].

Building on the tabular model TabPFN [14], prior-data-fitted networks have been specialized for causal effects: CausalPFN [2], Do-PFN [18], and the CausalFM framework [9]. CausalPFN is trained on simulated data-generating processes under ignorability and returns calibrated intervals. When a table exceeds its context window, it retrieves context rows by a weak-effect rule rather than reading all rows, and it is not built for instrumental-variable settings. This dependence on a bounded context, and the gap between the simulated prior and real campaign data, motivate the breakdown we study.

Two recent non-foundation methods address the hard regimes directly. The Q-Learner writes the treated-to-control outcome ratio as a product of two odds ratios and is the most consistent method under low conversion [10]. GP-CATE models each arm with a Gaussian process and gives calibrated intervals when the control arm is small, where standard X-Learner intervals under-cover [4].

3 Method and Timeline

CausalPFN is used as an in-context estimator: a sample of rows is given as context, and the effect is read off for new units with

no training. We vary three axes: context size on large data, the control-arm share down to a few-placebo setting, and the base conversion rate. Along each axis we record where the model’s ranking advantage over the classical core disappears.

CausalPFN-Rank keeps the weights frozen and acts in two places. On the input side, it replaces the default context with a treatment-balanced sample and averages predictions over several retrieved contexts, compared against that default rather than a random one. On the output side, it adds a light re-ranking step: a ranking calibrator, a lower-confidence-bound score from the model’s uncertainty, and a ratio score for low conversion. Table 1 lists the methods and their roles.

Table 1
Methods and their role in the study.

Method	Role
CausalPFN	Foundation anchor
CausalPFN-Rank	Proposed training-free fix
S/T/X/DR + boosting	Tuned classical core
Causal Forest	Forest baseline
BCF	Bayesian baseline
DragonNet, DESCN	Neural baselines
Q-Learner	Specialist: low conversion
GP-CATE	Specialist: small control

The project runs over four weeks. Week 1 builds the harness, reproduces CausalPFN’s uplift example as a smoke test, wires up the baselines, and runs the first untuned comparison. Week 2 builds the three axes and adds the specialists on their regimes. Week 3 implements and ablates the two parts of CausalPFN-Rank. Week 4 runs the final comparison with confidence intervals and the efficiency summary, then writes up results and limitations.

Limitations

Qini has high variance, so we use bootstrap confidence intervals, at least ten seeds, and a primary metric fixed in advance. CausalPFN already retrieves context by a weak-effect rule, so the context part of CausalPFN-Rank is measured against that default, not a random baseline. The model sees only a bounded context window while gradient boosting trains on all rows; we report this and treat the context budget as part of the method. Real randomized data allow only ranking evaluation, while semi-synthetic data give effect accuracy but come from causal models close to the model’s training prior, which is itself the subject of H1. We subsample large tables and train no foundation model.

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